

41. First Missing Positive

Return smallest positive integer not in nums

$O(n \log n)$ Sort First int that isn't $i, i+1$
 $1, 2, 3$

$O(1)$

$A = [1, 2, 4]$ res bound to $[1..4]$

$[1..n+1]$

$O(n)$ Hashset, try every $[1..n+1]$

Use Input Array As Memory

① Change all ≤ 0 numbers to $\text{len}(A) + 2$

② map value $\rightarrow$ value - 1 0 1 2 3

$v \in A \rightarrow$ index in A 3, 0, 6, 3

$3 \rightarrow 3-1 = 2$

$6 \rightarrow 6-1 = 5$ OOB

$3 \rightarrow 3-1 = 2$ already negative

A negative value at $A[i]$ signifies that

0 1 2 3

3, 0, -6, 3

1 2 3 4 $[1.. \text{len}(A)]$

③ Use the array as a hashset

Iterate from $[1.. \text{len}(A) + 1]$ 1 2 3 4

candidates

$[0.. \text{len}(A)]$ 0 1 2 3

indices

until we find a positive number